MESA SCHOOL OF BUSINESS

Data Foundations · Pre-Read

Session 3: Nykaa

Big Mastery Question: Retention has budget to save 500 customers. Which 500, and what's the rule that picks them?

Before you come to class

Read this once, slowly. Last session you taught Sheets to make a yes/no decision on ONE condition at a time. This session is about counting, summing and averaging based on conditions — the exact toolkit behind almost every 'how many' and 'how much' question a founder ever asks. Every table below shows real column letters and row numbers, same as always.

The company: Nykaa is India's biggest beauty and personal care platform, selling skincare, makeup and haircare online. Every customer has a history: how many times they've ordered, how much they've spent, whether they're a Luxe (premium) member. Your job this session: build the rule that picks the 500 customers retention should spend its budget saving — and that rule needs you to count and sum with conditions, not just add everything up blindly.

0. The Dataset You'll Be Working With

Nykaa's customer table has 9 columns (customer_id, signup_date, city, orders_count, total_spend, last_order_date, category_pref, is_luxe_member, email_optin). Here's a 6-column, 6-row slice:

	A	C	D	E	G	H
1	customer_id	city	orders_count	total_spend	category_pref	is_luxe_member
2	N3001	Bengaluru	5	12500	Skincare	TRUE
3	N3002	Mumbai	1	900	Makeup	FALSE
4	N3003	Bengaluru	8	21300	Skincare	TRUE
5	N3004	Delhi	2	1800	Haircare	FALSE
6	N3005	Mumbai	6	15400	Makeup	TRUE
7	N3006	Bengaluru	1	650	Haircare	FALSE

Column A = customer_id, C = city, D = orders_count, E = total_spend, G = category_pref, H = is_luxe_member. Columns B, F, I exist in the real sheet but aren't shown here.

1. COUNTIF

Real world reference:

You already met COUNTA — it counts every filled box, no questions asked. COUNTIF is pickier: it's like counting only the kids in a classroom wearing a red T-shirt. You give it a group to look at AND a condition to match, and it only counts the ones that pass.

In Nykaa's world:

Before you can decide anything about Bengaluru customers, you need to know how many there even are. COUNTIF looks down the city column and counts only the rows where the city is "Bengaluru" — ignoring Mumbai, Delhi, and everywhere else.

Syntax:

```
=COUNTIF(range, criteria)  Example: =COUNTIF(C2:C7, "Bengaluru")
```

See it in the table:

	A	C
1	**customer_id**	**city**
2	N3001	Bengaluru
3	N3002	Mumbai
4	N3003	Bengaluru
5	N3004	Delhi
6	N3005	Mumbai
7	N3006	Bengaluru
8	**Count →**	**=COUNTIF(C2:C7,"Bengaluru") = 3**

2. COUNTIFS

Real world reference:

Now add a second rule to your classroom check: count only kids wearing a red T-shirt AND sitting in the front row. COUNTIFS is COUNTIF's big sibling — it checks as many conditions as you give it, and only counts a row if EVERY single one is true.

In Nykaa's world:

You want Bengaluru customers who are ALSO Luxe members — Nykaa's premium tier. That's two conditions at once: city = Bengaluru AND is_luxe_member = TRUE. COUNTIFS checks both for every row and only counts the ones that satisfy both.

Syntax:

```
=COUNTIFS(range1, criteria1, range2, criteria2, ...)  Example:
=COUNTIFS(C2:C7,"Bengaluru", H2:H7,TRUE)
```

See it in the table:

	A	C	H
1	**customer_id**	**city**	**is_luxe_member**
2	N3001	Bengaluru	TRUE
3	N3002	Mumbai	FALSE
4	N3003	Bengaluru	TRUE
5	N3004	Delhi	FALSE

6	N3005	Mumbai	TRUE
7	N3006	Bengaluru	FALSE
8	**Count →**		**=COUNTIFS(C2:C7,"Bengaluru",H2:H7,TRUE) = 2**

N3001 and N3003 are Bengaluru AND Luxe — that's 2. N3006 is Bengaluru but NOT Luxe, so COUNTIFS correctly skips it.

3. SUMIF

Real world reference:

Remember SUM — pushing piles of blocks together? SUMIF only pushes together the piles that match a condition first. Sort your blocks by color, then only add up the red ones.

In Nykaa's world:

The founder wants to know: how much total revenue has come from Bengaluru customers? SUMIF checks the city column for "Bengaluru" and, for every matching row, adds up the corresponding value from the total_spend column.

Syntax:

```
=SUMIF(range, criteria, sum_range)  Example: =SUMIF(C2:C7, "Bengaluru", E2:E7)
```

See it in the table:

	A	C	E
1	**customer_id**	**city**	**total_spend**
2	N3001	Bengaluru	12500
3	N3002	Mumbai	900
4	N3003	Bengaluru	21300
5	N3004	Delhi	1800
6	N3005	Mumbai	15400
7	N3006	Bengaluru	650
8	**Total →**		**=SUMIF(C2:C7,"Bengaluru",E2:E7) = 34450**

Notice the order of arguments: range to CHECK first (city), criteria second, and the range to ADD UP (total_spend) comes last.

4. SUMIFS

Real world reference:

Same idea as COUNTIFS — but now you're adding up numbers instead of counting rows, and you can stack as many conditions as you like.

In Nykaa's world:

You want total spend from Bengaluru customers who specifically prefer Skincare — a very specific slice for a category-targeted campaign. SUMIFS checks both city AND category_pref, and only adds up total_spend where both match.

Syntax:

```
=SUMIFS(sum_range, range1, criteria1, range2, criteria2, ...)  Example:
=SUMIFS(E2:E7, C2:C7,"Bengaluru", G2:G7,"Skincare")
```

See it in the table:

	A	C	G	E
1	**customer_id**	**city**	**category_pref**	**total_spend**
2	N3001	Bengaluru	Skincare	12500
3	N3002	Mumbai	Makeup	900
4	N3003	Bengaluru	Skincare	21300
5	N3004	Delhi	Haircare	1800
6	N3005	Mumbai	Makeup	15400
7	N3006	Bengaluru	Haircare	650
8	**Total →**			**=SUMIFS(E2:E7,C2:C7,"Bengaluru",G2:G7,"Skincare") = 33800**

Careful — in SUMIF the sum_range comes LAST. In SUMIFS it comes FIRST. That flip trips up almost everyone the first time; it's worth double-checking every time you switch between the two.

5. AVERAGEIF

Real world reference:

Same as AVERAGE from Session 1 — sharing candies fairly — but now you only share among the kids who match one condition first.

In Nykaa's world:

What does a typical Mumbai customer spend on Nykaa? AVERAGEIF checks the city column for "Mumbai," then averages the total_spend only across matching rows — ignoring every other city.

Syntax:

```
=AVERAGEIF(range, criteria, average_range)  Example: =AVERAGEIF(C2:C7, "Mumbai",
E2:E7)
```

See it in the table:

	A	C	E
1	**customer_id**	**city**	**total_spend**
3	N3002	Mumbai	900
6	N3005	Mumbai	15400
—	**Average →**		**=AVERAGEIF(C2:C7,"Mumbai",E2:E7) = 8150**

Only Mumbai's two customers (rows 3 and 6 in the full sheet) get averaged — the other four rows are simply skipped.

6. AVERAGEIFS

Real world reference:

The AVERAGEIF version of COUNTIFS/SUMIFS — average with as many conditions stacked together as you need.

In Nykaa's world:

You want the average spend of Mumbai customers who are ALSO Luxe members — a much more specific, high-value segment than "all Mumbai customers." AVERAGEIFS checks both conditions before including a row in the average.

Syntax:

```
=AVERAGEIFS(average_range, range1, criteria1, range2, criteria2, ...)  Example:
=AVERAGEIFS(E2:E7, C2:C7,"Mumbai", H2:H7,TRUE)
```

See it in the table:

	A	C	H	E
1	**customer_id**	**city**	**is_luxe_member**	**total_spend**
3	N3002	Mumbai	FALSE	900
6	N3005	Mumbai	TRUE	15400
—	**Average →**			**=AVERAGEIFS(E2:E7,C2:C7,"Mumbai",H2:H7,TRUE) = 15400**

N3002 is Mumbai but NOT Luxe, so it's excluded — only N3005 qualifies, so the 'average' here is just its one value.

7. Locking Criteria Ranges with $

Real world reference:

Recall the signpost from Session 1 that always points to the same front door no matter where you're standing. When you build ONE formula and want to reuse it for many different cities, you need the DATA range nailed down like that signpost — while the city you're asking about is free to change from row to row.

In Nykaa's world:

You want a mini summary table listing every city with its customer count, built with ONE formula dragged down three times — not typed three separate times. Lock the data range ($C$2:$C$7) so it never shifts, while the city name (A2, A3, A4) is left relative so it updates for each row.

Syntax:

```
=COUNTIF($C$2:$C$7, A2)     ← data range locked, criteria cell free to move
```

See it in the table:

	A	B
1	**City**	**Customer Count**
2	Bengaluru	=COUNTIF($C$2:$C$7,A2) = 3

3	Mumbai	=COUNTIF($C$2:$C$7,A3) = 2
4	Delhi	=COUNTIF($C$2:$C$7,A4) = 1

This is a brand-new, separate mini-table — not part of the main customer sheet. The formula was typed once in B2, then autofilled down to B3 and B4.

8. Deriving AOV & Repeat Rate

Real world reference:

If you know a shop made ₹10,000 across 5 sales, you can instantly tell someone "the average sale was ₹2,000" — you're combining two numbers you already have into a new, more useful one. That's exactly what AOV (Average Order Value) and Repeat Rate are: not new Sheets functions, just useful ratios built from columns you already have.

In Nykaa's world:

AOV per customer tells you how much they spend PER ORDER, not just in total — a customer who spent ₹12,500 across 5 orders (₹2,500/order) is a very different shopper from one who spent the same across 1 order. Repeat Rate tells retention what share of the whole customer base has ordered more than once — the single number that defines whether Nykaa is retaining anyone at all.

Syntax:

```
AOV per customer:   = total_spend / orders_count          → =E2/D2 Repeat Rate:
= COUNTIF(range,">1") / COUNTA(range)  → =COUNTIF(D2:D7,">1")/COUNTA(D2:D7)
```

See it in the table:

	A	D	E	K
1	**customer_id**	**orders_count**	**total_spend**	**AOV**
2	N3001	5	12500	=E2/D2 = 2500
3	N3002	1	900	=E3/D3 = 900
4	N3003	8	21300	=E4/D4 = 2662.5
5	N3004	2	1800	=E5/D5 = 900
6	N3005	6	15400	=E6/D6 = 2566.67
7	N3006	1	650	=E7/D7 = 650

Repeat Rate across this sample: 4 of the 6 customers have orders_count > 1 (N3001, N3003, N3004, N3005) → =COUNTIF(D2:D7,">1")/COUNTA(D2:D7) = 4/6 = 66.7%. That single percentage is what retention will track every month.

Coming to class

You now know how to count, sum and average with conditions — and how to turn raw columns into metrics a founder actually cares about. In class, we'll build the full at-risk customer list and use AOV and Repeat Rate to decide exactly which 500 customers retention should save.